

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2023

Bachelor in Computer Applications

Course Title: **Probability & Statistics**

Code No: Semester: **III**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

11. Discuss the role of statistics in computer application.
12. Define Statistics? The following of telephone calls received at an exchange for 200 successive one-minute intervals are given below.
13. Define correlation. Calculate and analyze the correlation coefficient between the number of study hours and the number of sleeping hours of different students.
14. Define regression. The following table gives information on ages and cholesterol levels for sample of 10 men.
15. Define binomial distribution. During one stage in the manufacture of integrated circuit chips, a coating must be applied. If 70% of chips received a thick enough coating, find the probability that among 15 chips: i) at least 12 will have thick enough coatings ii) exactly 10 will have thick enough coatings
16. In measuring reaction time, a psychologist estimates that the standard deviation is 0.95 sec, how large a sample of measurements must be taken in order to be 95% confident that the error his estimate of mean will not exceed 0.01 second?
17. Define sampling. A population consists of the four numbers 1, 3, 4, 8. (1) Write down all possible sample size of two without replacement. (ii) Show that sample mean is an unbiased estimate of population mean.

Group C

Attempt any TWO questions [2x10=20]

18. An analysis of monthly wages paid to workers in two firms A and B belonging to the same industry gives the following results:
19. The length of human pregnancies from conception to birth approximates a normal distribution with a mean of 266 days and a standard deviation of 16 days. (i) What probability of all pregnancies will last between 240 and 270 days? (ii) What probability of all pregnancies will last more than 275 days? (iii) What probability of all pregnancies will last less than 272 days?

20. Define ANOVA. Given data, using one-way ANOVA test the hypothesis that means of three samples are equal